

Milestone 2: Feature Engineering & Baseline Modelling

1. Objective

The goal of Milestone 2 was to implement a baseline scanner identification system using handcrafted features and classical machine learning models. Key objectives were:

- Extract discriminative features from scanner images (noise patterns, frequency, texture).
 - Visualize differences between scanners.
 - Train baseline models and evaluate performance.
 - Identify limitations of handcrafted features.
-

2. Feature Engineering

2.1 Patch Extraction

- Each image (256×256) was divided into overlapping patches.
- Initial patch size: 32×32, stride 8.
- Later, denser patches (e.g., 24×24, stride 4) were tested to increase sample size.

Purpose: Patch-level analysis increases the number of training samples and allows capturing local scanner noise patterns.

2.2 Handcrafted Features

For each patch, the following features were extracted:

1. **Noise Patterns:**
 - Residual noise obtained via high-pass filtering.
 - Captures sensor-specific fixed pattern noise.
 2. **Frequency Domain Features:**
 - Fast Fourier Transform (FFT) to analyse frequency spectrum of noise.
 - Includes mean, variance, and energy of frequency components.
 3. **Texture Descriptors:**
 - Local Binary Patterns (LBP) to capture local texture differences caused by scanner electronics.
 4. **Wavelet Features:**
 - Discrete Wavelet Transform (DWT, Haar) applied to patches.
-

3. Baseline Model Training

3.1 Initial Attempt

- **Model:** XGBoost classifier
- **Result:** ~50%
- **Observation:** Mixture of flatfield and Wikipedia datasets masked the scanner specific patterns.

3.2 Improved Approach

- **Model:** Random Forest classifier
 - **Hyperparameters:**
 - `n_estimators=600, max_depth=30, min_samples_leaf=1, max_features='sqrt', class_weight='balanced'`
 - **Pipeline Enhancements:**
 1. Patch-level feature extraction (noise + FFT + LBP + DWT)
 2. Random Forest training on patch features
 3. **Image-level majority voting:** Each test image's patch predictions were aggregated to produce a single image-level scanner prediction.
-

5. Evaluation

5.1 Metrics

- **Accuracy:** Image-level accuracy reached **~90.9%**
- **Classification Report & Confusion Matrix:** Computed for each scanner class

5.2 Visualization

- Random correct and incorrect predictions were visualized:
 - Correct samples show accurate scanner noise identification.
 - Misclassified samples mostly correspond to challenging noise patterns or patch boundary artifacts.
-

6. Limitations

1. **Handcrafted features:** Capture most of the discriminative noise patterns, but may miss subtle scanner-specific variations.
2. **Document images:** Cannot be used directly for training without preprocessing, as content dominates scanner noise features.

3. **Accuracy ceiling:** With current data and handcrafted features, image-level accuracy plateaued around ~91%.
-

7. Summary & Key Learnings

- Patch-based feature extraction and Random Forest with image-level voting significantly improved scanner identification accuracy.
- Handcrafted features (noise, FFT, LBP, DWT) successfully captured scanner-specific patterns.
- Dataset limitations and document content interfere with model performance.
- Future improvements may include:
 - More flat-field images per scanner
 - Advanced preprocessing for document images
 - Deep learning approaches with transfer learning